X-Checks Automation

Include & Exclude Account Handling

Representation in FIP Text Output and EBX Excel Workbook

This briefing documents every mechanism used in the FIP Validation Rule text output and the EBX Cross Checks publication file to represent included and excluded accounts or reporting units. It covers observed patterns, current match/mismatch status, and the implications for the X-Checks comparison process.

1. Background

The X-Checks process compares formulas and variables extracted from two sources: the EBX "cross checks all" Excel sheet (which defines the rules) and the FIP Validation Rule text output (which records what the EPM system has actually applied). Both sources can restrict a check to a subset of reporting units (Included/Excluded RUs) or exclude specific account types from the calculation. These restrictions are encoded differently in each source.

The EBX workbook uses three dedicated columns: **Included RUs**, **Excluded RUs**, and **Exclude Account Type**. The FIP text output embeds the restriction directly inside the variable name within the formula string, using a variety of notations.

2. FIP Text Output — Exclusion Representation

FIP encodes exclusions as suffixes appended to the variable name inside VAL_YTD() calls. Four distinct mechanisms have been identified, with the account-type exclusion appearing in six different syntactic variants.

2.1 Mechanism 1 — EXN_ Account Prefix

Certain FS Account codes carry an **EXN_** prefix, indicating that they hold aggregated data for excluded entities. These accounts appear in the variable section of the FIP block alongside normal accounts. The exclusion is structural — it is baked into the account code itself rather than being a formula annotation.

32 distinct EXN_ codes are present in the test file:

EXN_11100	EXN_11121	EXN_11122	EXN_11131	EXN_11132	EXN_11140
EXN_11221	EXN_11222	EXN_12100	EXN_12251	EXN_12252	EXN_12260
EXN_13100	EXN_14100	EXN_15100	EXN_15210	EXN_21100	EXN_21220
EXN_21220ff	EXN_21231	EXN_21231ff	EXN_21232	EXN_21233	EXN_21236
EXN_21240	EXN_21250	EXN_22100	EXN_22209	EXN_22210	EXN_23100
EXN_24100	EXN_25100	EXN_25210	—	—	—

Example (LR048_17): FIP variables include EXN_11121, EXN_11122, EXN_12100, EXN_21100, EXN_22100 etc. alongside the main FS accounts.

2.2 Mechanism 2 — Account Type Exclusion Suffix

When an account type is excluded, FIP appends a suffix directly to the variable name inside the VAL_YTD() call. The same concept appears in six syntactic variants:

Suffix Form	Example	Observed In
excl.acc.type=2	VAL_YTD(IAN_00051excl.acc.type=2)	AS447_09, AS448_09
excl.acc.type:2	VAL_YTD(IAN_00023exl.acc.type:2)	AS130_00
exl.acc.type2	VAL_YTD(LIN_00355ffexl.acc.type2)	AL167_00 (typo: "exl" not "excl")
excl.acct.type 2	VAL_YTD(EXN_21231ff excl.acct.type 2)	EBX variables section
excl. acc. type = 2	VAL_YTD(LIN_00380 excl. acc. type = 2 ToM L07)	Formula text
excl. acc. type:1,4	VAL_YTD(LIN_00380 excl. acc. type:1,4 TOM L09)	Multi-type exclusion
excl.acc.type2 (no sep)	VAL_YTD(LIN_00380 excl.acc.type2 TOM L08)	Compressed form

All variants express the same concept: exclude account type 2 (Affiliated). The inconsistency in spacing, punctuation and abbreviation ("acc" vs "acct", "excl" vs "exl") originates from the EPM system and must be normalised before comparison.

2.3 Mechanism 3 — Posting Level Exclusion (excl PL)

A single observed case uses **excl PL** notation to exclude a specific posting level:

```
VAL_YTD(23151ffexclPL10) = 0
```

This appears in X-Check AL109_70. The EBX file has Excluded RUs = CON_30104 for this check, but there is no direct column mapping between a CON_ code and a posting-level exclusion. This is an isolated case.

2.4 Mechanism 4 — Text Annotations (excl. CH, excluding SXX)

Some formulas carry free-text exclusion annotations within the variable name:

Annotation	Meaning	Example Context
excl. CH / excl CH units	Excludes Swiss consolidation units	AL166_00, L123_00 context text

Annotation	Meaning	Example Context
excluding S02	Excludes segment S02	VAL_YTD(10102 excluding S02)
excluding S31	Excludes segment S31	VAL_YTD(10102 excluding S31)
excl. 2 - Aff.	Excludes affiliated (Group 2)	SLSTLSAS_11520n-lifeLOBsexcl.2-Aff
excl.2 -Aff	Same — compressed variant	Variable name suffix

3. EBX Workbook — Exclusion Columns

The EBX "cross checks all" sheet has four columns relevant to inclusion/exclusion. All are populated at row level (one row per account/sub-account combination per X-Check), though in practice the values are consistent across all rows of the same X-Check.

3.1 Included RUs

93 rows across 89 distinct X-Checks carry an Included RUs value. This specifies which consolidation unit the check applies to — only that RU is in scope. The 11 distinct values observed are:

CON_ Code	Row Count	X-Check Count	Entity
CON_LFLOB	72 rows	~40 X-Checks	Life France Line of Business
CON_ZIC	11 rows	~10 X-Checks	Zurich Insurance Company
CON_RR	6 rows	6 X-Checks	Reinsurance entity
CON_ARG2	2 rows	2 X-Checks	Argentina entity
CON_DE	12 rows	12 X-Checks	Germany entity
CON_ZIP	1 row	1 X-Check	Zurich Insurance Plc
CON_CH2	2 rows	2 X-Checks	Swiss entity variant
CON_GI17	1 row	1 X-Check	General Insurance 2017
CON_PC&BBA;	1 row	1 X-Check	P&C; / B&A; entity
CON_201000	2 rows	2 X-Checks	Entity code 201000
CON_123SII	1 row	1 X-Check	SII entity
CON_ZICZR	1 row	1 X-Check	ZIC Zurich Reinsurance
CON_ZLIC	1 row	1 X-Check	Zurich Life Insurance Company

Effect on FIP formula: None. In every tested case, X-Checks with Included RUs produce identical formulas in both FIP and EBX. The CON_ code is informational metadata only — it does not change the formula text.

3.2 Excluded RUs

171 rows across 30 distinct X-Checks carry an Excluded RUs value. This specifies which consolidation unit is out of scope for the check. 25 distinct CON_ codes are observed:

CON_ Code	Rows	X-Checks	Formula Match	Notes
CON_400069	50 rows	1 X-Check	Match	A142_00
CON_DIR	29 rows	Multiple	Match	Directorate entity
CON_ZIP	26 rows	Multiple	Match	Zurich Insurance Plc
CON_30104	10 rows	1 X-Check	MisMatch	AL109_70 — FIP adds "exclPL10"
CON_ARG2	7 rows	Multiple	Match	Argentina entity
CON_RR	7 rows	Multiple	Match	Reinsurance entity
CON_51012	6 rows	Multiple	Match	—
CON_SWISS	4 rows	1 X-Check	Match	LS601_00
CON_SISTER	4 rows	1 X-Check	Match	L019_00
CON_201001	4 rows	1 X-Check	Match	LS169_00
CON_CH	2 rows	2 X-Checks	Match	AL166_00, L123_00

CON_ Code	Rows	X-Checks	Formula Match	Notes
CON_FA	3 rows	1 X-Check	Match	R007_00
CON_DE	1 row	1 X-Check	Match	S123_70
Others	~30 rows	Multiple	Match	12 further CON_ codes

Effect on FIP formula: None in almost all cases. The single exception is AL109_70 (CON_30104) where FIP appends "exclPL10" — a posting-level exclusion that has no direct column equivalent in EBX. All other Excluded RUs produce identical formulas.

3.3 Exclude Account Type

130 rows across 50 distinct X-Checks carry an Exclude Account Type value. This is the most impactful column — it directly affects the formula structure in FIP. Five distinct values are observed:

EBX Value	Row Count	Formula Match	Notes
2 - Affiliated	118	MisMatch	FIP appends excl.acc.type=2 suffix; may also flip operator from + to –
2 - Affiliated, 6 - Unit linked	5	Match	FIP does NOT add notation for this combination — formulas match as-is
2 - Affiliated, 9 - IC Difference	3	Match	FIP does NOT add notation for this combination — formulas match as-is
1 - 3rd, 4 - Linked	2	TBC	Not yet observed in FIP output — needs further investigation
1-3rd, 4-Linked, 6-Unit linked	2	TBC	Not yet observed in FIP output — needs further investigation

3.4 Exclude Z-Core

This column is present in the EBX workbook but contains no data in either test file. No action required.

4. Detailed Formula Impact — Exclude Account Type = 2 - Affiliated

This is the only case where a formula mismatch is consistently produced by an exclusion column. Two sub-patterns are observed depending on the formula structure:

4.1 Pattern A — Suffix only, no operator change

When the excluded account is the *only* variable (or all variables share the same operator), FIP appends the exclusion suffix to the variable name without changing the operator:

```
EBX: ABS(VAL_YTD(LIN_00355ff)) >= CONST(1000,'USD','E')
FIP: ABS(VAL_YTD(LIN_00355ffexl.acc.type2)) >= CONST(1000,'USD','E')
```

X-Check: AL167_00 | Account: LIN_00355^OAN_00261 | Excl: 2 - Affiliated

4.2 Pattern B — Suffix + operator flip + variable reorder

When the excluded account is one of multiple variables in an addition formula, FIP restructures the expression: the excluded variable moves to the end, its operator changes from **+** to –, and the exclusion suffix is appended:

```
EBX: ABS(VAL_YTD(IAN_00051) + VAL_YTD(S12301ffToM349)) <= CONST(5,'USD','E')
FIP: ABS(VAL_YTD(S12301ffToM349) - VAL_YTD(IAN_00051excl.acc.type=2)) <= CONST(5,'USD','E')
```

X-Check: AS447_09 | Excluded account: IAN_00051 | Excl: 2 - Affiliated

Summary of observed mismatching X-Checks:

X-Check	EBX Excl. Type	Excluded Acct	Change Type	FIP Variable Name	Pattern
AL167_00	2 - Affiliated	LIN_00355ff	Suffix only	LIN_00355ffexl.acc.type2	Pattern A
AS130_00	2 - Affiliated	IAN_00023	Suffix + op flip	IAN_00023exl.acc.type:2	Pattern B
AS447_09	2 - Affiliated	IAN_00051	Suffix + op flip	IAN_00051excl.acc.type=2	Pattern B
AS448_09	2 - Affiliated	SN_13105	Suffix + op flip	SN_13105excl.acc.type=2	Pattern B

4.3 FIP Notation Variants for Account Type 2

All of the following have been observed in the FIP text for account type 2 exclusions. They represent the same concept and must be normalised to a single canonical form:

FIP Notation	Example X-Check / Context
excl.acc.type=2	AS447_09, AS448_09
exl.acc.type:2	AS130_00 (note: typo "exl" not "excl")
exl.acc.type2	AL167_00 (note: typo "exl" not "excl")
excl.acct.type 2	EXN_ variable blocks
excl. acc. type = 2	LIN_00380 formula text
excl.acc.type2	LIN_00380 compressed formula text

5. Current Match Status Summary

Exclusion Type	EBX Value	X-Checks	Status	Notes
Included RUs	All	89	Match	CON_ codes are metadata; no formula change
Excluded RUs (most)	All	29	Match	CON_ codes are metadata; no formula change
Excluded RUs (AL109_70)	CON_30104	1	MisMatch	FIP adds "exclPL10" — no EBX column maps to posting level
Excl. Acc. Type 2-Aff.	2 - Affiliated	~4	MisMatch	FIP appends suffix + may flip operator
Excl. Acc. Type 2+6	2-Aff, 6-UL	~5	Match	FIP does not add notation for this combination
Excl. Acc. Type 2+9	2-Aff, 9-IC	~3	Match	FIP does not add notation for this combination
Excl. Acc. Type 1,4	1-3rd, 4-Lnk	TBC	Unknown	Not yet observed in FIP test data
SLST/IFRSN prefix mismatch	n/a	~6	MisMatch	Structural naming difference; separate from exclusion handling

6. Implications for the Comparison Process

6.1 What must be built

To achieve matching formulas for **Exclude Account Type = 2 - Affiliated**:

(a) FIP Normalisation

The 6 syntactic variants of account type 2 exclusion in FIP must be normalised to a single canonical form before comparison. The recommended canonical form is **excl.acc.type=2** (the most common form observed).

(b) EBX Formula Enhancement

For accounts with Exclude Account Type = "2 - Affiliated", the EBX formula builder must:

- Append the canonical exclusion suffix to the variable name
- Move the excluded variable to the end of the VAL_YTD expression
- Change its operator from + to – (Pattern B cases)
- Leave the operator unchanged for Pattern A cases (single variable or ABS structure)

6.2 What does NOT need to be built

The following require no code changes as they either already produce matching formulas or are out of scope:

- Included RUs — purely informational; no formula impact
- Excluded RUs (all except AL109_70) — purely informational; no formula impact
- Excl. Acc. Type 2 + 6 and 2 + 9 — FIP already matches EBX without change
- SLST / IFRSN prefix mismatches — structural naming difference; separate investigation required

6.3 Open questions

- **AL109_70 / excl PL**: No EBX column maps to posting-level exclusions. Is this a gap in the EBX file, or is posting-level exclusion out of scope?
- **Excl. Acc. Type 1-3rd, 4-Linked**: Not yet observed in FIP test data. What notation does FIP use for account types 1 and 4?
- **Pattern A vs Pattern B determination**: The rule for when to flip the operator (Pattern B) vs leave it unchanged (Pattern A) is not yet fully defined. Further examples are needed to confirm whether it is determined by the

Operator (X-Check Term) column value or the formula structure.

7. Data Sources

Source	Filename	Notes
EBX workbook	20251205 EPM X-Checks - Original - Copy.xlsx	Sheet: cross checks all
FIP text file	20251205 FIP X-Checks - Original.txt	Full Validation Rule export
Comparison output	20260515_095332_X-Checks Comparison.xlsx	Generated by new X-Checks app

Analysis date: 2026-05-15 | Branch: v0.3-X-Checks | Generated by X-Checks automation tooling